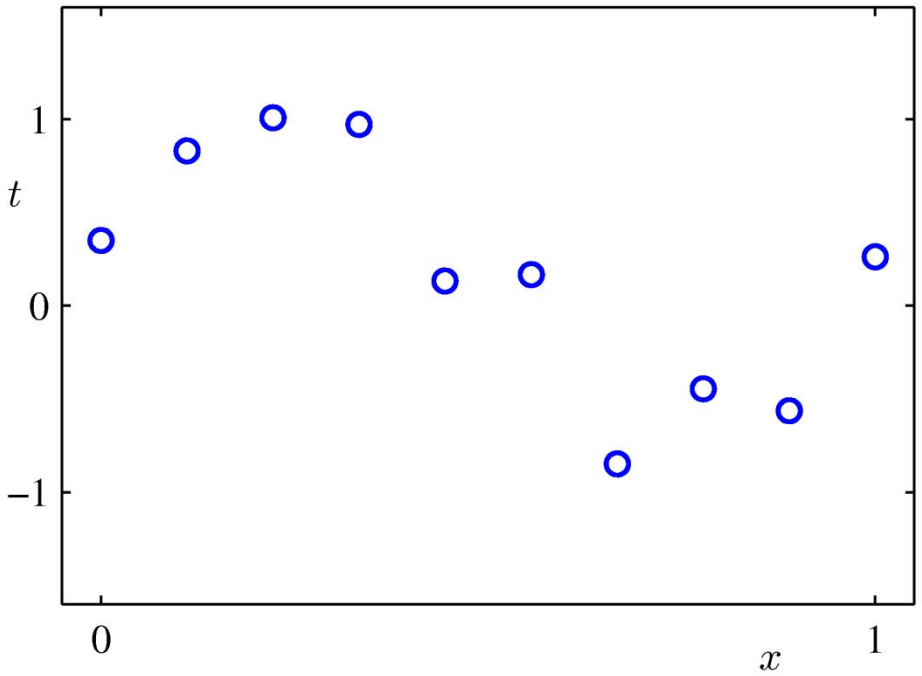
Overfitting, Generalization, Cross-Validation

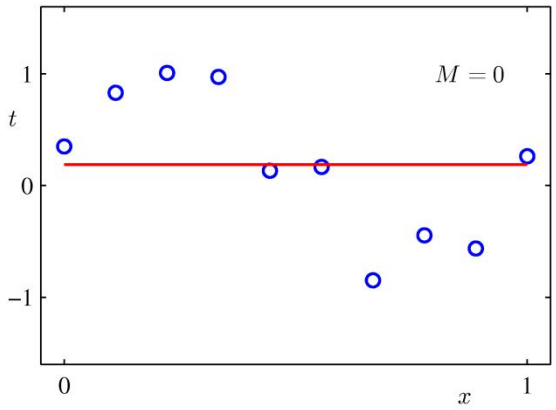
Overfitting vs Generalization

- **Of Which order polynomial will be best for the data?**

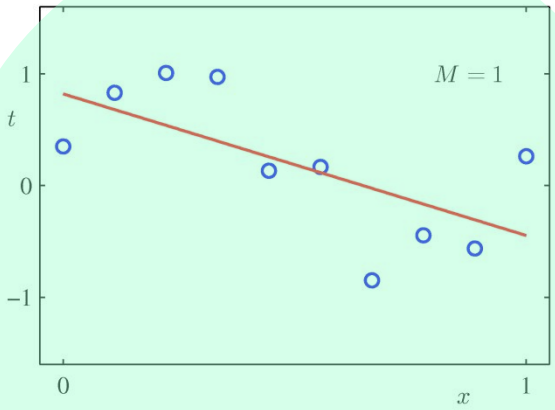


Overfitting vs Generalization

- **Of Which order polynomial will be best for the data?**
  - The model which has the least error as much as possible



0th order polynomial regression

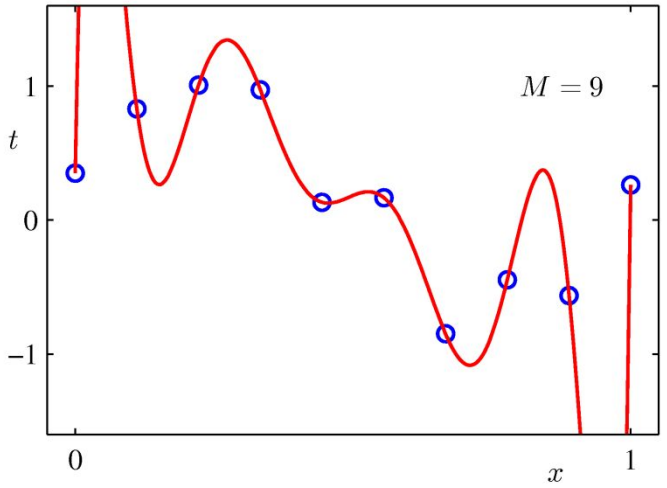


1st order polynomial regression

This is better because it has less error

Overfitting vs Generalization

- **Of Which order polynomial will be best for the data?**
  - What about this?



9th order polynomial regression

  - This may be the BEST because the error is ZERO!!

Do you agree with this?

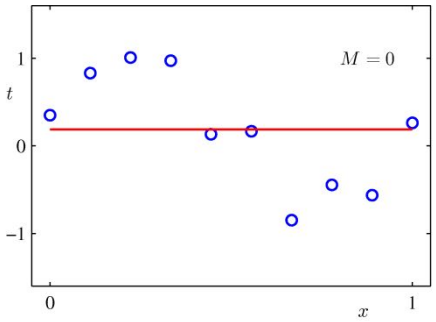
Overfitting vs Generalization

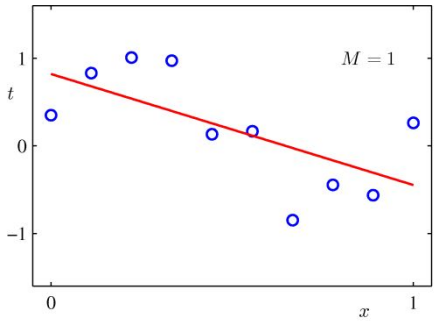
- **What is the purpose of Machine Learning?**

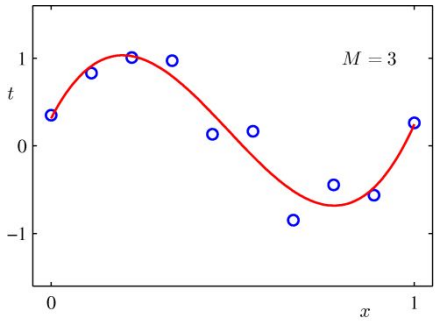
Learning the given data as exactly as possible

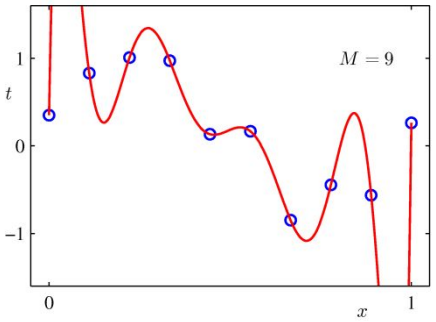
vs

Predict the unknown data as exactly as possible based on the given data



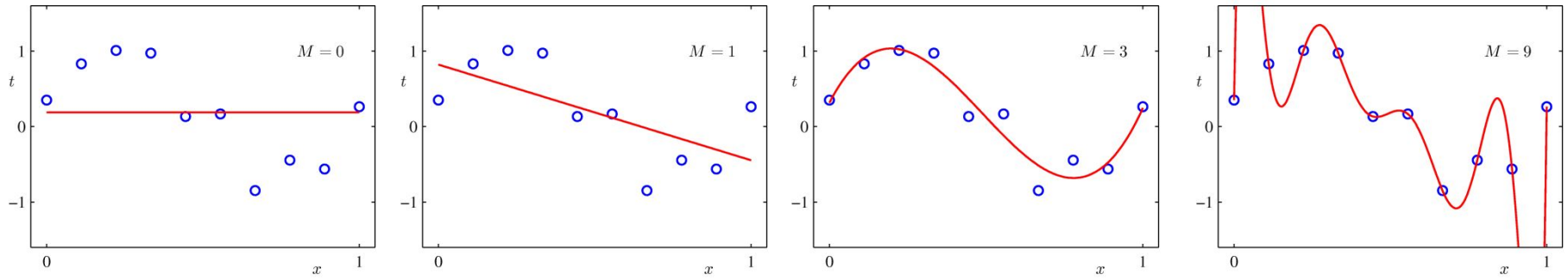






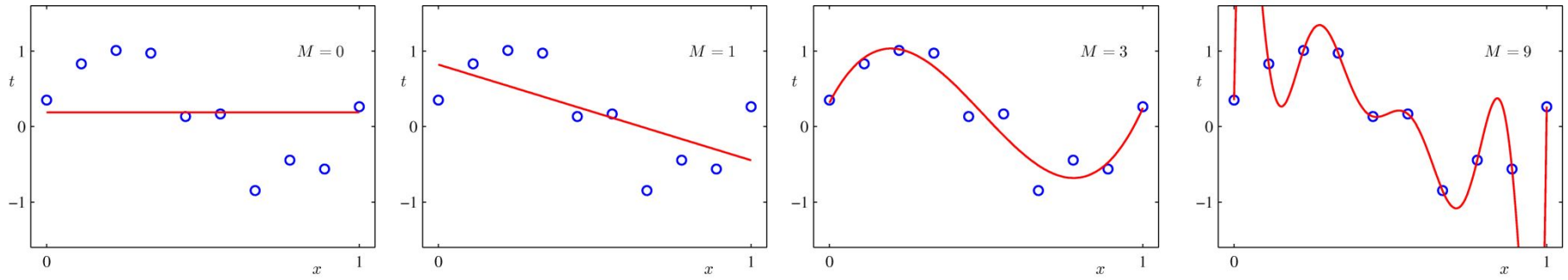
Overfitting vs Generalization

- **Then, which one looks best?**
  - As the order (M) increases,
    - the complexity of model increases
  - As the complexity of model increases,
    - the model can more exactly learn the given data
    - However, the prediction accuracy does not necessarily increase



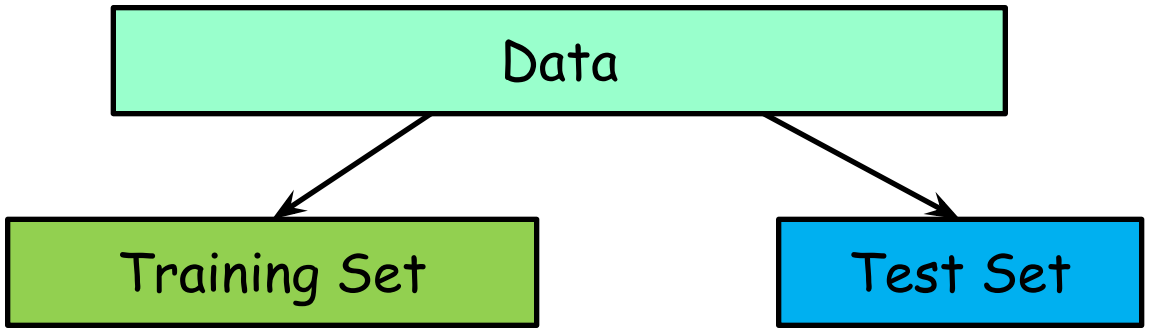
Model Evaluation

- **Which model is best?**
  - You may try several approaches and need to choose one
  - You may try several different parameters of a model and need to choose one
- **Model Evaluations**
  - Based on Training & Testing data set
  - Cross-Validation



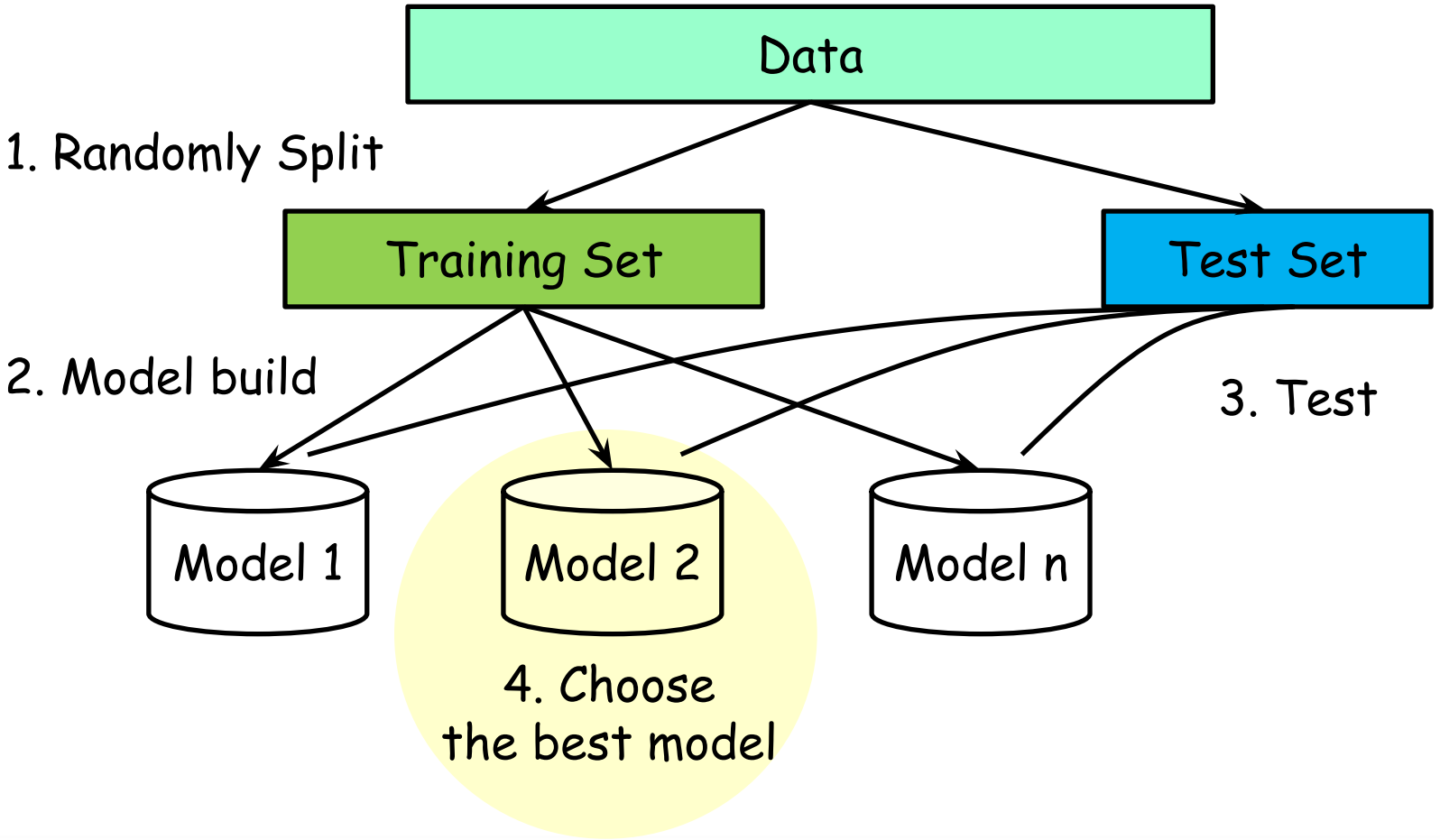
Training Set and Test Set

- **How to choose a good model**
  - Divide the given data into TRAINING set and TEST set
    - Training set and Test set should NOT overlap each other!!
    - Both need to be independent as much as possible
  - With Training set, build various models
  - With Test set, evaluate each model
  - Choose the model which shows the best performance with Test set



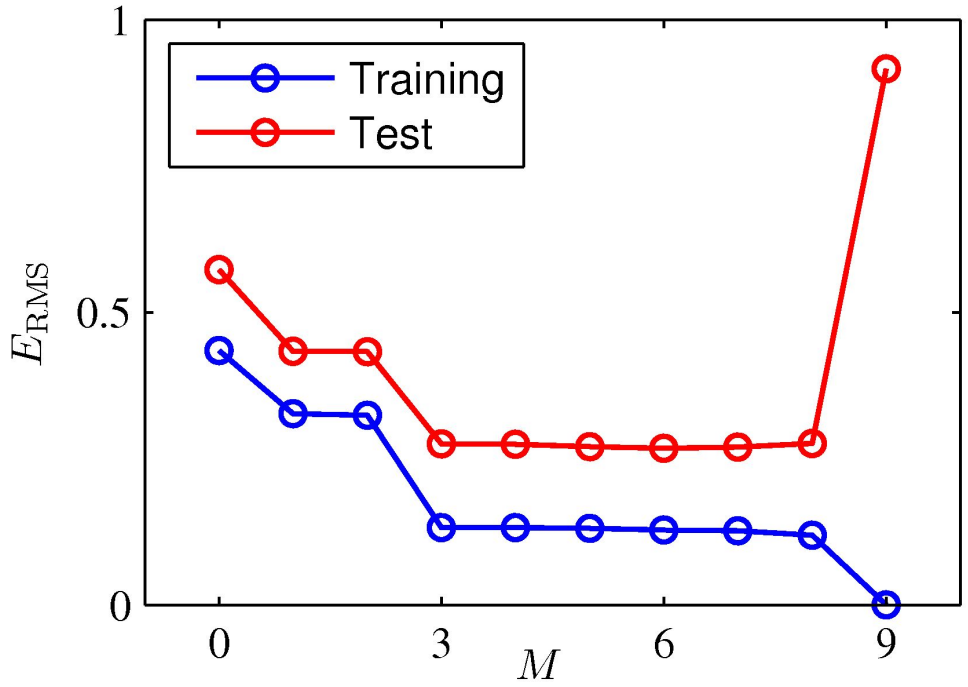
Training Set and Test Set

- **How to choose a good model**



Training Set and Test Set

- **Performance Graph**



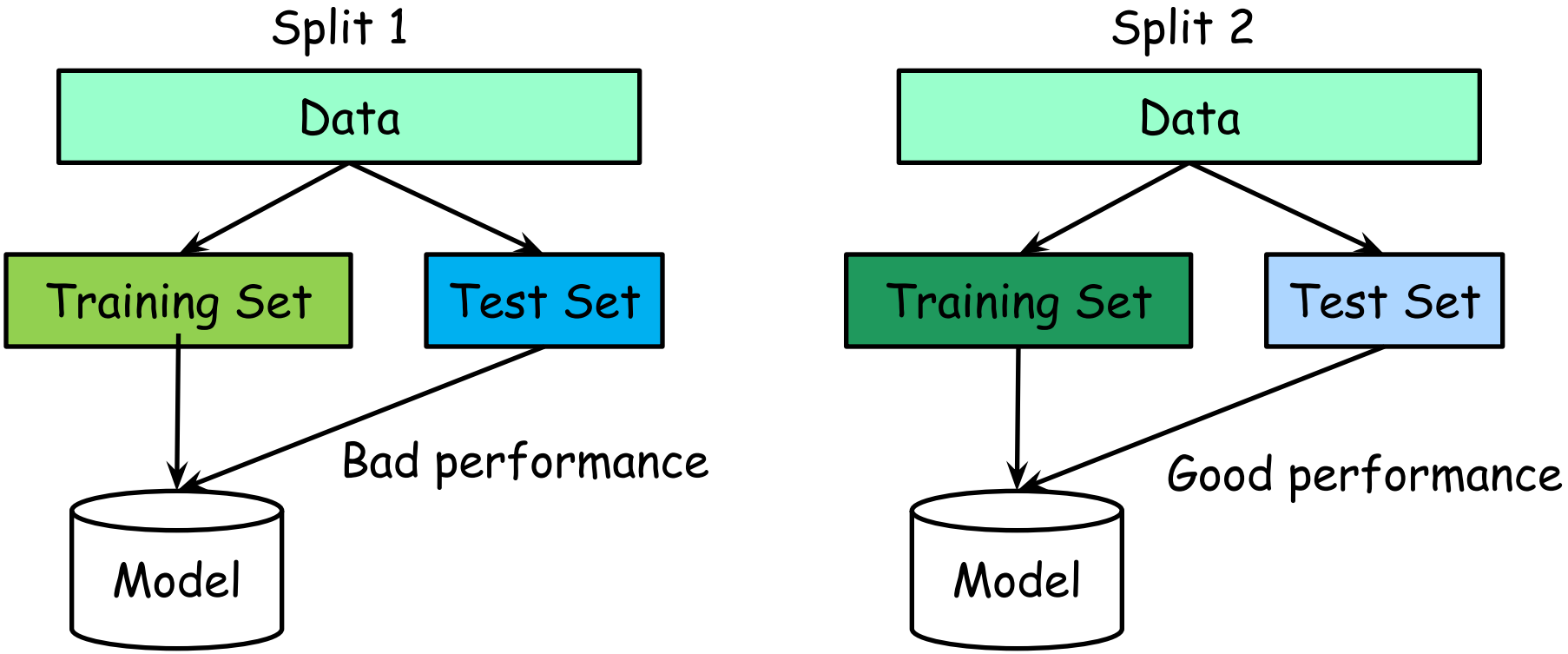
Training Set and Test Set

- **Size of Test set**
  - 50~30% of given data
- **Advantage**
  - Simple & easy
- **Disadvantage**
  - Test set is not used for modeling building. Waste of data
  - Data is randomly split
    - Evaluation can be significantly different depending on data split

=> Any good idea?

Training Set and Test Set

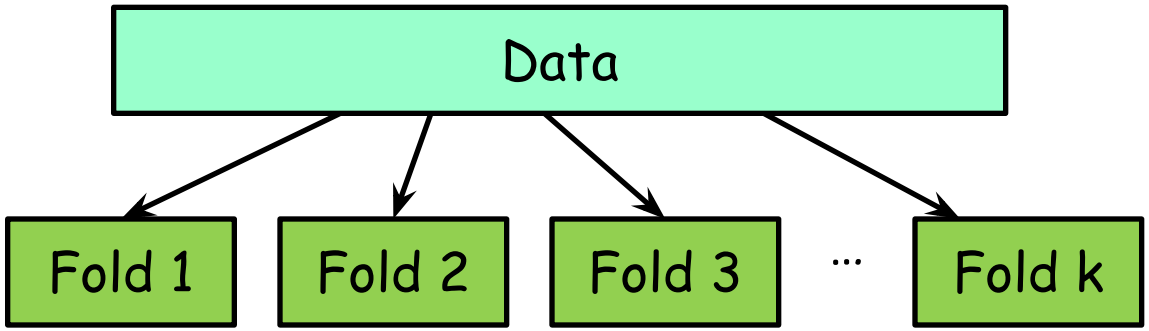
- **Data Waste & Random Split**



Cross Validation

- **In order to reduce statistical variance**
  - Usually, k-fold cross validation is widely used

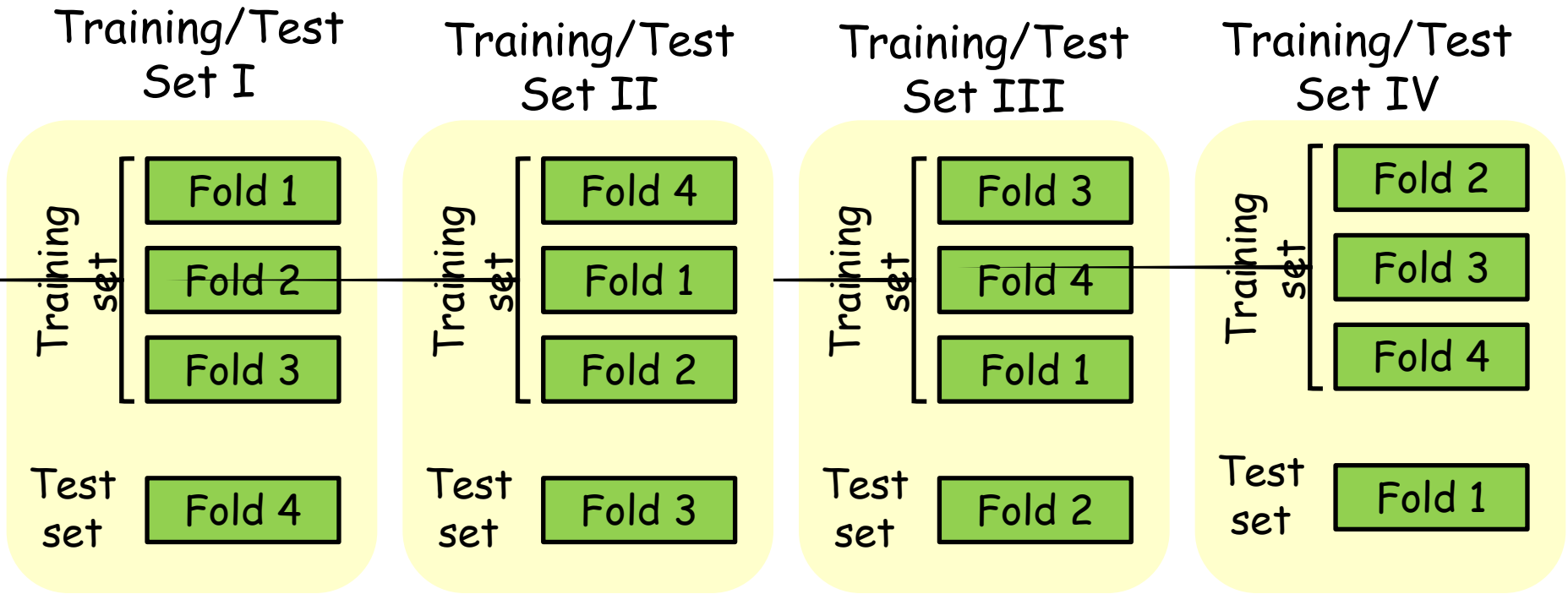
- **K-fold Cross Validation**
  - Split given data into *K* folds
  - Folds should not overlap with each other



  - Compose *k-1* training set and 1 test set with k folds

Cross Validation

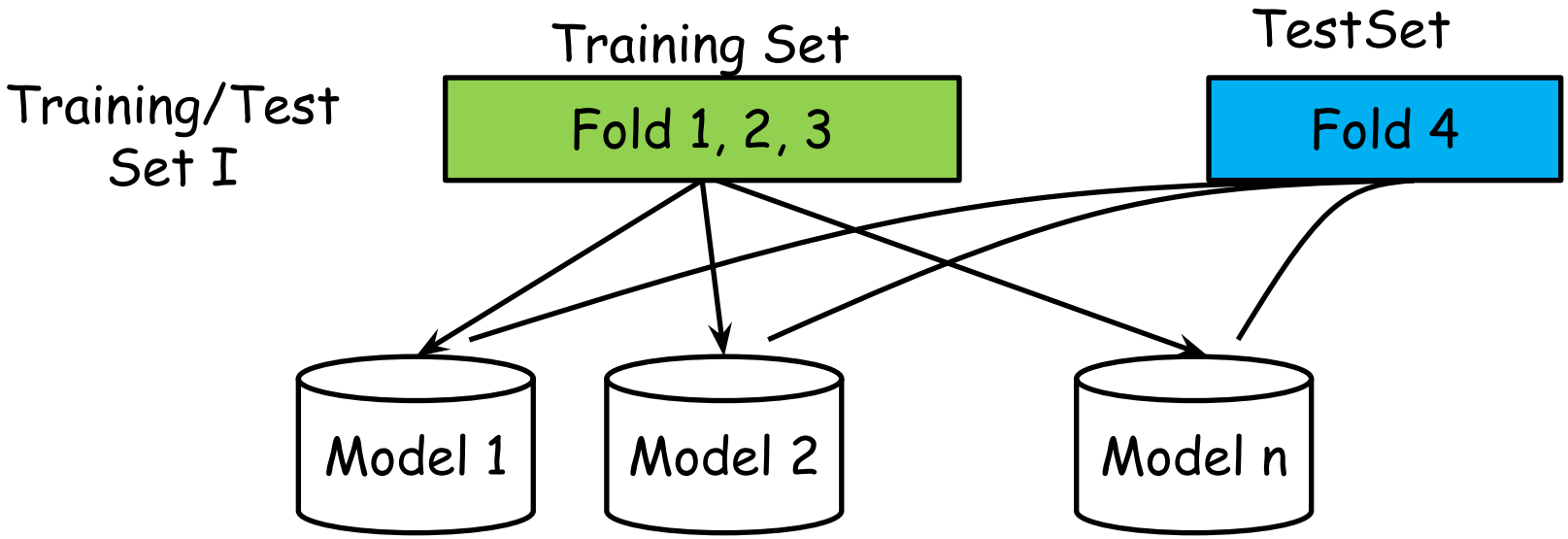
- **Example: 4 fold cross validation**



Choose a model by the average performance of 4 sets

Cross Validation

- **Example: 4 fold cross validation**

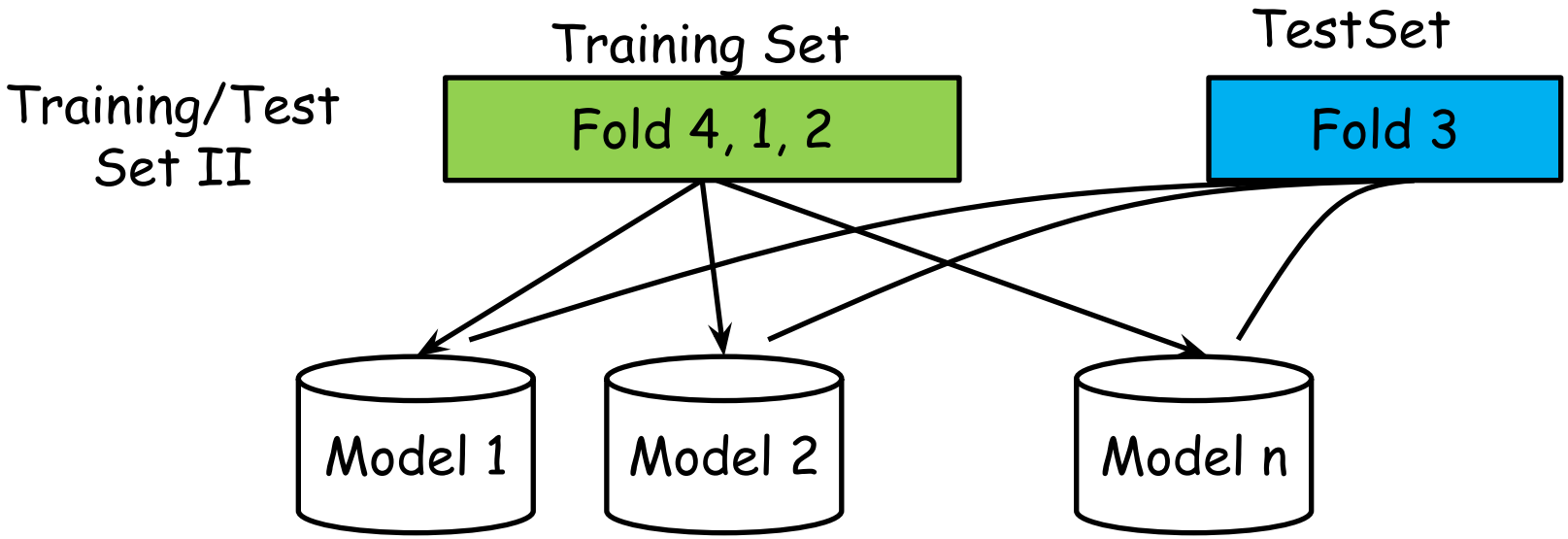


Set I	3	4	…	5
Set II			…	
Set III			…	
Set IV			…	

Performance

Cross Validation

- **Example: 4 fold cross validation**

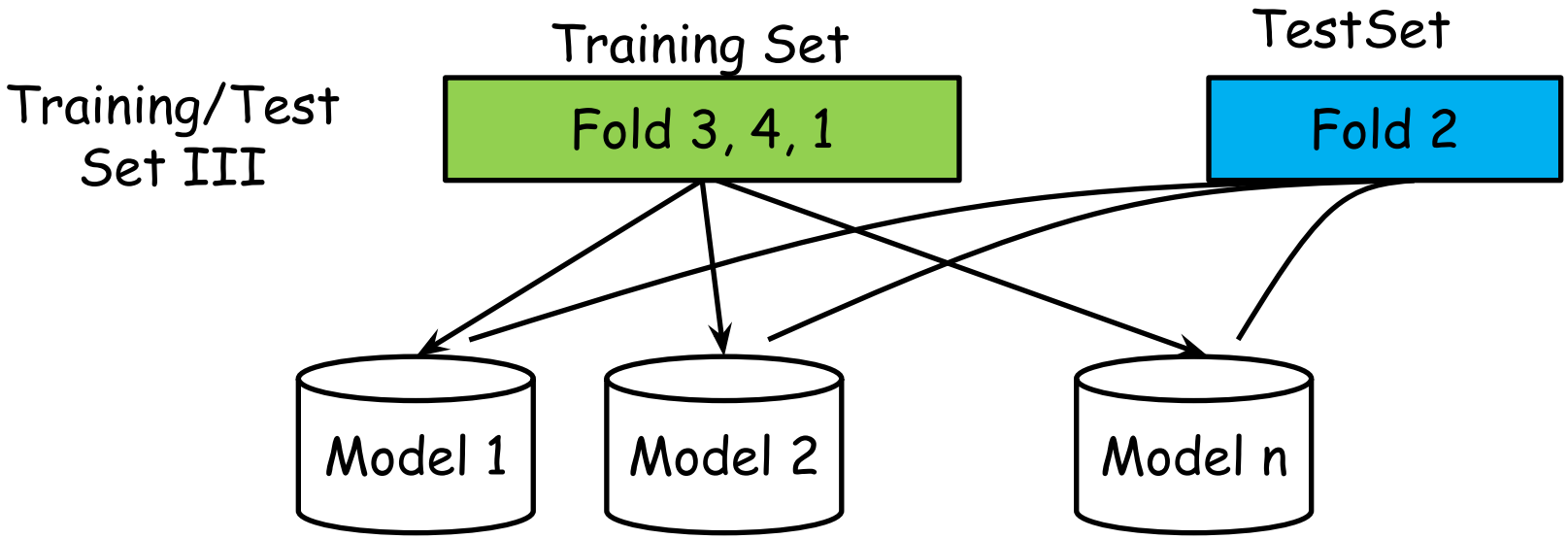


Set I	3	4	…	5
Set II	4	1	…	2
Set III			…	
Set IV			…	

Performance

Cross Validation

- **Example: 4 fold cross validation**

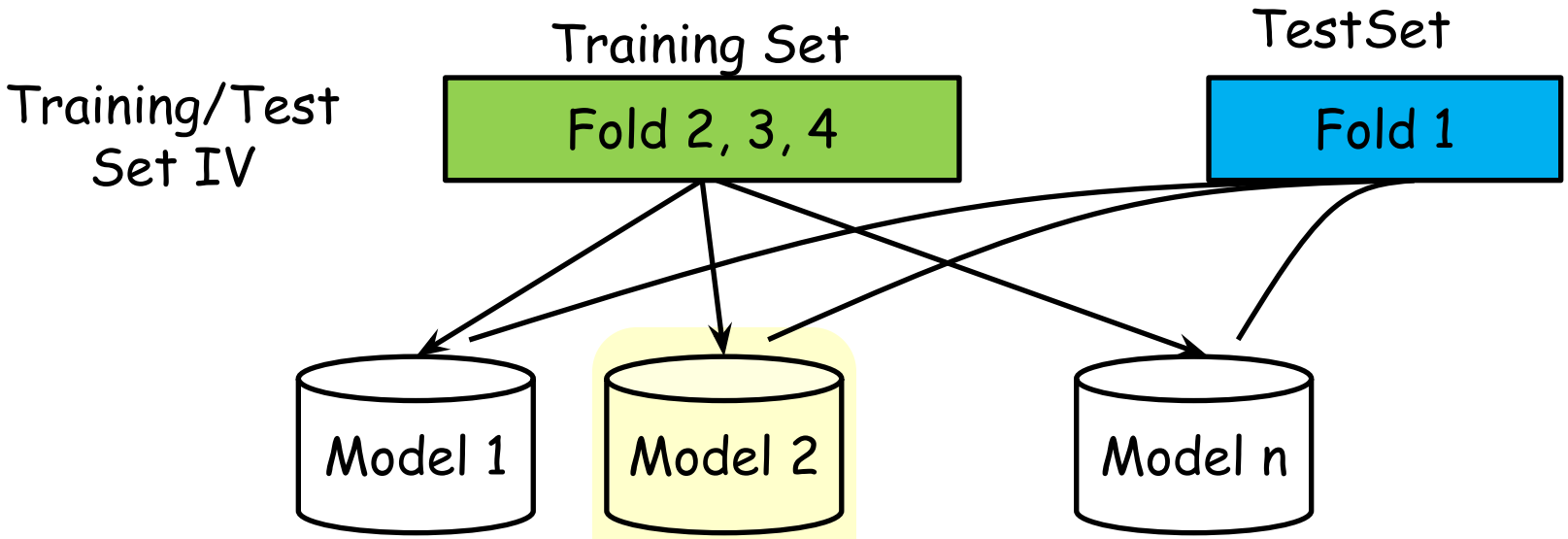


Set I	3	4	…	5
Set II	4	1	…	2
Set III	5	2	…	3
Set IV			…	

Performance

Cross Validation

- **Example: 4 fold cross validation**



Set I	3	4	…	5
Set II	4	1	…	2
Set III	5	2	…	3
Set IV	4	3	…	2

Performance

Cross Validation

- **Summary**
  - The data set is divided into *k* subsets, and the holdout method is repeated *k* times.
  - Each time, one of the *k* subsets is used as the test set and the other *k-1* subsets are put together to form a training set.
  - Then the average error across all *k* trials is computed.
  - The variance is reduced as *k* is increased.
- **Advantage**
  - Less dependent on how the data gets divided.
  - Every data point gets to be in a test set exactly once, and gets to be in a training set k-1 times.
- **Disadvantage**
  - Time !!

Variation

- **No fixed folds and Random Division**
  - Regular CV: Split into folds and Shuffling folds
  - No split into folds and Randomly dividing the data into a test and training set k different times.

- **LOOCV (Leave-one-out Cross Validation)**
  - Extreme case of *k*-fold cross validation
  - If data size is *n*, set $k = n$
  - Every data point except one is used for training and the remaining one is used for testing
  - Repeat this *n* times